

RATNAGIRI, India

WMO: 431100

Lat: 16.983N

Lon: 73.333E

Elev: 220

StdP: 14.58

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>64.3 | (c)<br>66.0 | (d)<br>44.1                   | (e)<br>42.9 | (f)<br>80.7 | (g)<br>47.8 | (h)<br>49.6 | (i)<br>80.0 | (j)<br>14.3           | (k)<br>81.2 | (l)<br>12.1 | (m)<br>80.9 | (n)<br>2.3            | (o)<br>90 | (p)<br>0.301 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 10.5                   | 93.9            | 73.7 | 92.6 | 74.4 | 91.4 | 74.9 | 81.6                | 87.7 | 80.9 | 86.9 | 80.5 | 86.3 | 5.0                  | 320  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.0                            | 156.9 | 84.9 | 79.3 | 153.4 | 84.3 | 78.9 | 151.0 | 83.9 | 45.6          | 87.9 | 44.9 | 87.1 | 44.3 | 86.5 | 86.0              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min  | Max  | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 12.6              | 11.1 | 9.4  | DB                         | 60.0 | 99.6 | 2.6                | 3.2                                                | 58.1      | 101.9 | 56.6       | 103.7 | 55.1       | 105.5 | 53.2       | 107.9 |
|                   |      |      | WB                         | 53.1 | 83.0 | 2.0                | 1.2                                                | 51.7      | 83.8  | 50.6       | 84.6  | 49.5       | 85.2  | 48.0       | 86.1  |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 80.8   | 77.2  | 78.0  | 80.7  | 83.4  | 85.3  | 81.9  | 80.1  | 79.9  | 80.0  | 82.2  | 81.9  | 79.4  |
|                                                                                  | DBStd       | 3.22   | 2.89  | 3.06  | 2.63  | 1.94  | 1.81  | 3.04  | 1.99  | 1.57  | 1.71  | 2.36  | 2.27  | 2.73  |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50       | 11255  | 843   | 784   | 950   | 1001  | 1095  | 956   | 935   | 926   | 899   | 1000  | 956   | 910   |
|                                                                                  | CDD65       | 5781   | 378   | 365   | 485   | 551   | 630   | 506   | 470   | 461   | 449   | 535   | 506   | 445   |
|                                                                                  | CDH74       | 59718  | 3236  | 3267  | 4971  | 6687  | 8327  | 5560  | 4486  | 4206  | 4044  | 5599  | 5256  | 4079  |
| CDH80                                                                            | 21393       | 1222   | 1270  | 1900  | 2745  | 3957  | 1872  | 819   | 610   | 866   | 2151  | 2295  | 1686  |       |
| Wind                                                                             | WSAvg       | 3.5    | 3.2   | 3.2   | 3.2   | 3.7   | 4.1   | 4.1   | 4.9   | 4.2   | 2.7   | 2.6   | 2.9   | 3.3   |
| Precipitation                                                                    | PrecAvg     | 112.2  | 0.0   | 0.0   | 0.0   | 0.1   | 1.2   | 32.2  | 36.8  | 23.5  | 13.7  | 3.5   | 1.7   | 0.4   |
|                                                                                  | PrecMax     | 154.4  | 0.6   | 0.1   | 0.2   | 0.5   | 7.1   | 51.9  | 55.8  | 53.5  | 32.6  | 16.8  | 11.6  | 5.7   |
|                                                                                  | PrecMin     | 72.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 15.0  | 17.6  | 5.2   | 4.6   | 0.0   | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 22.8   | 0.1   | 0.0   | 0.0   | 0.1   | 1.9   | 9.3   | 9.7   | 13.0  | 7.7   | 4.0   | 2.7   | 1.3   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 92.5  | 94.5  | 95.4  | 93.9  | 93.9  | 92.5  | 86.7  | 86.1  | 89.1  | 95.3  | 94.8  | 93.2  |
|                                                                                  |             | MCWB   | 68.5  | 67.4  | 71.7  | 76.3  | 79.0  | 79.7  | 79.6  | 78.8  | 79.1  | 74.1  | 72.3  | 70.2  |
|                                                                                  | 2%          | DB     | 90.2  | 91.3  | 91.5  | 91.3  | 92.2  | 90.2  | 85.4  | 84.8  | 86.9  | 93.2  | 93.1  | 91.3  |
|                                                                                  |             | MCWB   | 68.6  | 69.4  | 73.6  | 77.5  | 79.5  | 79.6  | 79.3  | 78.6  | 78.3  | 75.1  | 72.4  | 70.0  |
|                                                                                  | 5%          | DB     | 88.5  | 89.2  | 89.4  | 90.0  | 91.1  | 88.7  | 84.5  | 83.9  | 85.5  | 91.3  | 91.6  | 89.8  |
|                                                                                  |             | MCWB   | 68.4  | 70.2  | 74.6  | 77.7  | 79.2  | 79.6  | 79.1  | 78.4  | 78.0  | 76.0  | 72.5  | 69.5  |
|                                                                                  | 10%         | DB     | 86.6  | 87.2  | 87.7  | 88.9  | 90.2  | 87.1  | 83.6  | 83.1  | 84.3  | 89.0  | 90.1  | 88.1  |
|                                                                                  |             | MCWB   | 68.7  | 70.8  | 75.2  | 77.6  | 78.9  | 79.5  | 78.7  | 78.1  | 77.7  | 77.1  | 72.7  | 69.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 75.3  | 77.2  | 80.1  | 81.8  | 82.7  | 82.1  | 80.9  | 80.4  | 80.9  | 81.9  | 79.7  | 77.4  |
|                                                                                  |             | MCDB   | 83.6  | 84.4  | 87.4  | 88.2  | 89.4  | 87.5  | 84.6  | 84.0  | 85.8  | 88.1  | 87.3  | 85.4  |
|                                                                                  | 2%          | WB     | 73.6  | 75.6  | 79.1  | 80.7  | 81.6  | 81.3  | 80.1  | 79.5  | 79.9  | 80.8  | 78.4  | 75.7  |
|                                                                                  |             | MCDB   | 83.0  | 83.6  | 86.1  | 87.4  | 88.5  | 86.5  | 83.8  | 83.0  | 84.2  | 86.9  | 85.8  | 84.6  |
|                                                                                  | 5%          | WB     | 72.5  | 74.3  | 78.2  | 79.9  | 80.9  | 80.7  | 79.6  | 79.0  | 79.1  | 79.9  | 77.6  | 74.5  |
|                                                                                  |             | MCDB   | 82.5  | 82.9  | 85.1  | 86.8  | 87.9  | 85.9  | 83.2  | 82.5  | 83.3  | 85.9  | 85.2  | 83.6  |
|                                                                                  | 10%         | WB     | 71.3  | 73.0  | 77.3  | 79.1  | 80.3  | 80.1  | 79.1  | 78.4  | 78.5  | 79.1  | 76.6  | 73.1  |
|                                                                                  |             | MCDB   | 81.8  | 82.2  | 84.1  | 86.2  | 87.4  | 85.1  | 82.5  | 81.8  | 82.5  | 85.0  | 84.4  | 82.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 19.8  | 19.0  | 15.6  | 12.1  | 10.5  | 7.8   | 5.6   | 5.8   | 8.5   | 13.8  | 18.3  | 19.8  |
|                                                                                  |             | MCDBR  | 21.8  | 21.8  | 17.7  | 13.1  | 11.4  | 10.3  | 6.8   | 6.9   | 10.5  | 18.0  | 20.2  | 21.0  |
|                                                                                  |             | MCWBR  | 10.0  | 10.1  | 8.2   | 5.3   | 3.8   | 3.7   | 3.5   | 3.3   | 4.2   | 7.1   | 8.4   | 9.3   |
|                                                                                  | 5% WB       | MCDBR  | 19.6  | 18.7  | 13.6  | 11.4  | 10.8  | 8.7   | 6.2   | 6.2   | 8.7   | 13.1  | 15.5  | 17.9  |
| MCWBR                                                                            |             | 9.3    | 9.2   | 6.9   | 5.2   | 4.4   | 4.3   | 3.7   | 3.4   | 4.4   | 6.0   | 6.4   | 7.9   |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.456 | 0.459 | 0.535 | 0.592 | 0.570 | 0.631 | 0.667 | 0.608 | 0.537 | 0.587 | 0.528 | 0.476 |
|                                                                                  | taud        |        | 2.153 | 2.167 | 1.984 | 1.879 | 1.961 | 1.840 | 1.755 | 1.886 | 2.080 | 1.905 | 2.024 | 2.114 |
|                                                                                  | Ebn at Noon |        | 262   | 268   | 251   | 237   | 239   | 223   | 215   | 230   | 247   | 230   | 238   | 251   |
|                                                                                  | Edh at Noon |        | 46    | 48    | 59    | 65    | 60    | 67    | 73    | 64    | 53    | 61    | 52    | 47    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1660  | 1914  | 2122  | 2246  | 2149  | 1356  | 1112  | 1238  | 1485  | 1641  | 1605  | 1567  |
|                                                                                  | RadStd      |        | 50    | 45    | 63    | 54    | 123   | 158   | 171   | 99    | 143   | 87    | 81    | 42    |

Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | +0.27 | N/A     | +1.89  | +0.74   | +0.61 | +0.67 | N/A         | N/A   | N/A   | N/A   |
| Regional (3 neighbors) | +0.34 | N/A     | +1.04  | +0.72   | +0.52 | +0.49 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page